

1) If $A = \begin{bmatrix} a_1 & a_2 \\ b_1 & b_2 \end{bmatrix}$ is a stochastic matrix and $v = [v_1, v_2]$ is a probability vector, show that VA is also a probability vector.

By data;

$$a_1 + a_2 = 1 \quad \& \quad b_1 + b_2 = 1$$

and also, $v_1 + v_2 = 1$

Now consider, $VA = [v_1, v_2] \begin{bmatrix} a_1 & a_2 \\ b_1 & b_2 \end{bmatrix} = [v_1 a_1 + v_2 b_1, v_1 a_2 + v_2 b_2]$

we have to prove that; $(v_1 a_1 + v_2 b_1) + (v_1 a_2 + v_2 b_2) = 1$

$$\begin{aligned} \text{LHS} &= v_1 a_1 + v_2 b_1 + v_1 a_2 + v_2 b_2 \\ &= v_1 (a_1 + a_2) + v_2 (b_1 + b_2) \\ &= v_1 (1) + v_2 (1) \\ &= v_1 + v_2 \\ &= 1 \end{aligned}$$

$\therefore$ Thus VA is a probability vector

2) Prove with reference to two second order stochastic matrices that their product is also a stochastic matrix.

Let $A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$ and $B = \begin{bmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{bmatrix}$ be two stochastic matrices;

Hence we have;

$$a_{11} + a_{12} = 1 \quad \& \quad b_{11} + b_{12} = 1$$

$$a_{21} + a_{22} = 1 \quad \& \quad b_{21} + b_{22} = 1$$

consider;

$$AB = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \begin{bmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{bmatrix} =$$

$$\begin{bmatrix} a_{11}b_{11} + a_{12}b_{21} & a_{11}b_{12} + a_{12}b_{22} \\ a_{21}b_{11} + a_{22}b_{21} & a_{21}b_{12} + a_{22}b_{22} \end{bmatrix}$$

we have to prove that;

$$(a_{11}b_{11} + a_{12}b_{21}) + (a_{11}b_{12} + a_{12}b_{22}) = 1 \rightarrow (1)$$

$$(a_{21}b_{11} + a_{22}b_{21}) + (a_{21}b_{12} + a_{22}b_{22}) = 1 \rightarrow (2)$$

4) F in eqn (1); LHS

$$a_{11}b_{11} + a_{12}b_{21} + a_{11}b_{12} + a_{12}b_{22}$$

$$= a_{11}(b_{11} + b_{12}) + a_{12}(b_{21} + b_{22})$$

$$= a_{11}(1) + a_{12}(1)$$

$$= a_{11} + a_{12}$$

$$= 1 \text{ RHS}$$

in eqn (2); LHS

$$a_{21}b_{11} + a_{22}b_{21} + a_{21}b_{12} + a_{22}b_{22}$$

$$= a_{21}(b_{11} + b_{12}) + a_{22}(b_{21} + b_{22})$$

$$= a_{21}(1) + a_{22}(1)$$

$$= a_{21} + a_{22}$$

$$= 1 \text{ RHS}$$

$\therefore$ Thus AB is a stochastic matrix.

3) If A is a square matrix of order n whose rows are each the same vector $a = [a_1, a_2, \dots, a_n]$ and if $v = (v_1, v_2, \dots, v_n)$ is a Probability vector; Prove that $VA = a$.

We have;

$$A = \begin{bmatrix} a_1 & a_2 & \dots & a_n \\ a_1 & a_2 & \dots & a_n \\ \vdots & \vdots & \ddots & \vdots \\ a_1 & a_2 & \dots & a_n \end{bmatrix} \quad \text{and } v_1 + v_2 + \dots + v_n = 1$$

Consider VA as a matrix product.

$$VA = [v_1, v_2, \dots, v_n] \begin{bmatrix} a_1 & a_2 & \dots & a_n \\ a_1 & a_2 & \dots & a_n \\ \vdots & \vdots & \ddots & \vdots \\ a_1 & a_2 & \dots & a_n \end{bmatrix}$$

$$= [v_1 a_1 + v_2 a_1 + \dots + v_n a_1, v_1 a_2 + v_2 a_2 + \dots + v_n a_2, \dots, v_1 a_n + v_2 a_n + \dots + v_n a_n]$$

$$= [a_1 (v_1 + v_2 + \dots + v_n), a_2 (v_1 + v_2 + \dots + v_n), \dots, a_n (v_1 + v_2 + \dots + v_n)]$$

$$= [a_1, a_2, a_3, \dots, a_n] = a \quad \text{Since } (v_1 + v_2 + \dots + v_n = 1)$$

Thus $VA = a$ as required.

4) Find the unique fixed probability vector of the regular stochastic matrix $A = \begin{bmatrix} 3/4 & 1/4 \\ 1/2 & 1/2 \end{bmatrix}$

we have to find $v = (x, y)$
where $x + y = 1$; such that $vA = v$.

$$\Rightarrow [x \ y] \begin{bmatrix} 3/4 & 1/4 \\ 1/2 & 1/2 \end{bmatrix} = [x \ y]$$

$$\left[\frac{3x}{4} + y/2 \quad x/4 + y/2 \right] = [x \ y]$$

$$x = \frac{3x}{4} + y/2 \rightarrow (1) \quad \& \quad y = x/4 + y/2 \rightarrow (2)$$

we can solve these 2 eqⁿ by using $y = 1 - x$

$$\frac{3x}{4} + y/2 = x \Rightarrow \frac{3x}{4} + \frac{(1-x)}{2} = x \Rightarrow \boxed{x = 2/3}$$

$$y = 1 - x \Rightarrow y = 1 - 2/3 \Rightarrow \boxed{y = 1/3}$$

Thus; $v = (2/3, 1/3)$ is the unique fixed probability vector.

5) find the unique fixed probability vector for the regular stochastic matrix $A = \begin{bmatrix} 0 & 1 & 0 \\ 1/6 & 1/2 & 1/3 \\ 0 & 2/3 & 1/3 \end{bmatrix}$

we have to find $v = (x, y, z)$

where $x + y + z = 1$; such that $vA = v$.

$$\text{i.e.; } [x \ y \ z] \begin{bmatrix} 0 & 1 & 0 \\ 1/6 & 1/2 & 1/3 \\ 0 & 2/3 & 1/3 \end{bmatrix} = [x \ y \ z]$$

$$\left[y/6 \quad x + y/2 + \frac{2z}{3} \quad \frac{y}{3} + \frac{z}{3} \right] = [x \ y \ z]$$

$$x = y/6 \quad \& \quad y = x + y/2 + \frac{2z}{3} \quad \& \quad z = \frac{y}{3} + \frac{z}{3}$$

we can solve these three equations by using $y = 1 - x - z$

$$y = 6x; \quad 6x + 3y + 4z = 6y; \quad y + z = 3z \Rightarrow y = 2z$$

$$y = 1 - x - z$$

$$y = 1 - y/6 - y/3$$

$$1 = y + y/6 + y/3$$

$$12y + 2y + 4y = 12$$

$$\frac{20y}{12} = 1$$

$$y = \frac{3}{5}$$

$$y = 6x$$

$$35 = 6x$$

$$x = \frac{1}{10}$$

$$y = 2z$$

$$\frac{3}{5} = 2z$$

$$z = \frac{3}{10}$$

Thus; $v = (\frac{1}{10}, \frac{3}{5}, \frac{3}{10})$ is the unique fixed prob vector.

6) with reference to stochastic A is $\begin{bmatrix} 3/4 & 1/4 \\ 1/4 & 1/4 \end{bmatrix}$ verify the property that the sequence A^2, A^3, A^4 whose rows are each the fixed probability vector.

$$A = \begin{bmatrix} 3/4 & 1/4 \\ 1/4 & 1/4 \end{bmatrix}$$

from problem (4) $v = (2/3, 1/3)$

$$\text{i.e.; } B = \begin{bmatrix} 2/3 & 1/3 \\ 2/3 & 1/3 \end{bmatrix} \Rightarrow \text{consider } A = \frac{1}{4} \begin{bmatrix} 3 & 1 \\ 1 & 1 \end{bmatrix}$$

$$\text{Now; } A^2 = \frac{1}{16} \begin{bmatrix} 3 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 3 & 1 \\ 1 & 1 \end{bmatrix} = \frac{1}{16} \begin{bmatrix} 11 & 5 \\ 4 & 2 \end{bmatrix} = \begin{bmatrix} 0.6875 & 0.3125 \\ 0.25 & 0.125 \end{bmatrix}$$

$$A^3 = \frac{1}{64} \begin{bmatrix} 3 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 11 & 5 \\ 4 & 2 \end{bmatrix} = \frac{1}{64} \begin{bmatrix} 43 & 21 \\ 14 & 7 \end{bmatrix} = \begin{bmatrix} 0.6718 & 0.3281 \\ 0.2187 & 0.1094 \end{bmatrix}$$

$$A^4 = \frac{1}{256} \begin{bmatrix} 3 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 43 & 21 \\ 14 & 7 \end{bmatrix} = \frac{1}{256} \begin{bmatrix} 171 & 85 \\ 57 & 28 \end{bmatrix} = \begin{bmatrix} 0.6679 & 0.3320 \\ 0.2227 & 0.1094 \end{bmatrix}$$

Each row of A^4 is approaching $v = (2/3, 1/3) = (0.67, 0.33)$

7) Find the unique fixed probability vector of the regular stochastic matrix

$$P = \begin{bmatrix} 0 & 1/3 & 1/4 & 1/4 \\ 1/3 & 0 & 1/4 & 1/4 \\ 1/3 & 1/3 & 0 & 0 \\ 1/3 & 1/3 & 0 & 0 \end{bmatrix}$$

We have to find $v = (a, b, c, d)$ where $a+b+c+d=1$ such that $VP = v$.

$$\Rightarrow [a, b, c, d] \begin{bmatrix} 0 & 1/3 & 1/4 & 1/4 \\ 1/3 & 0 & 1/4 & 1/4 \\ 1/3 & 1/3 & 0 & 0 \\ 1/3 & 1/3 & 0 & 0 \end{bmatrix} = [a, b, c, d]$$

$$\text{i.e.; } \left[\frac{b}{3} + \frac{c}{3} + \frac{d}{3}, \frac{a}{3} + \frac{c}{3} + \frac{d}{3}, \frac{a}{4} + \frac{b}{4}, \frac{a}{4} + \frac{b}{4} \right] = [a, b, c, d]$$

$$\Rightarrow \frac{1}{3}(b+c+d) = a \rightarrow (1)$$

$$\Rightarrow \frac{1}{3}(a+c+d) = b \rightarrow (2)$$

$$\Rightarrow \frac{1}{4}(a+b) = c \rightarrow (3)$$

$$\Rightarrow \frac{1}{4}(a+b) = d \rightarrow (4)$$

By using $b+c+d = 1-a$ & $a+c+d = 1-b$

eqn (1) becomes $\frac{1}{3}(1-a) = a \Rightarrow 1-a = 3a \Rightarrow 1-b = 3b$

$$1 = 3a \Rightarrow a = 1/3 \text{ \& } b = 1/3$$

$$\text{eqn (3)} \quad \frac{1}{4}\left(\frac{1}{3} + \frac{1}{3}\right) = c$$

$$= \frac{1}{4}\left(\frac{2}{3}\right) = c \Rightarrow \frac{2}{12} = c \Rightarrow c = \frac{1}{6}, d = \frac{1}{6}$$

Thus $v = (1/3, 1/3, 1/6, 1/6)$ is required unique fixed probability vector.

8) Show that (a, b) is a fixed point of the stochastic matrix $P = \begin{bmatrix} 1-b & b \\ a & 1-a \end{bmatrix}$ what is the associated fixed probability vector?

Hence write down the fixed probability vector of each of the following matrices.

$$P_1 = \begin{bmatrix} 1/3 & 2/3 \\ 1 & 0 \end{bmatrix}, P_2 = \begin{bmatrix} 1/2 & 1/2 \\ 2/3 & 1/3 \end{bmatrix}, P_3 = \begin{bmatrix} 7/10 & 3/10 \\ 8/10 & 2/10 \end{bmatrix}$$

let $x = (a, b)$ & consider the matrix product

$$\begin{aligned} xP &= [a, b] \begin{bmatrix} 1-b & b \\ a & 1-a \end{bmatrix} \\ &= [a(1-b) + ba, ab + b(1-a)] = [a, b] \end{aligned}$$

thus $xP = x$ $\therefore x = (a, b)$ is a fixed point of P .

also $v = (a/a+b, b/a+b)$ is the required fixed probability vector. Comparing P_1, P_2, P_3 with P we have

$$a=1, b=2/3; a=2/3, b=1/2; a=8/10, b=3/10.$$

$$a+b=5/3; a+b=7/6; a+b=11/10.$$

$$v = (a/a+b, b/a+b)$$

$$\text{Thus } v_1 = (3/5, 2/5); v_2 = (4/7, 3/7); v_3 = (8/11, 3/11).$$

are the required fixed probability vectors of P_1, P_2, P_3 in the respective order.

9) If $P_1 = \begin{bmatrix} 1-a & a \\ b & 1-b \end{bmatrix}$ and $P_2 = \begin{bmatrix} 1-b & b \\ a & 1-a \end{bmatrix}$ show that P_1, P_2 and $P_1 P_2$ are stochastic matrices.

In P_1 we have $(1-a) + a = 1$ & $b + (1-b) = 1$

In P_2 we have $(1-b) + b = 1$ & $a + (1-a) = 1$

Thus P_1, P_2 are stochastic matrices.

$$\begin{aligned} \text{Now } P_1 P_2 &= \begin{bmatrix} 1-a & a \\ b & 1-b \end{bmatrix} \begin{bmatrix} 1-b & b \\ a & 1-a \end{bmatrix} \\ &= \begin{bmatrix} (1-a)(1-b) + a^2 & (1-a)b + a(1-a) \\ b(1-b) + a(1-b) & b^2 + (1-b)(1-a) \end{bmatrix} = \begin{bmatrix} a_1 & b_1 \\ a_2 & b_2 \end{bmatrix}. \end{aligned}$$

We show that $a_1 + b_1 = 1$ & $a_2 + b_2 = 1$

$$\Rightarrow a_1 + b_1 = (1-a)(1-b) + (1-a)b + a^2 + a(1-a)$$

$$= (1-a) \{1-b+b\} + a \{a+1-a\}$$

$$= 1-a+a=1 \quad \therefore a_1 + b_1 = 1$$

$$\Rightarrow a_2 + b_2 = b(1-b) + b^2 + a(1-b) + (1-b)(1-a)$$

$$= b \{ (1-b) + b \} + (1-b) \{ a + (1-a) \}$$

$$= b + 1 - b \quad \therefore a_2 + b_2 = 1$$

thus $P_1 P_2$ is a stochastic matrix.

10) show that $P = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1/2 & 1/2 & 0 \end{bmatrix}$ is a regular stochastic

matrix. Also find the associated unique fixed probability vector.

$$\text{Consider } P^2 = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1/2 & 1/2 & 0 \end{bmatrix} \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1/2 & 1/2 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 1 \\ 1/2 & 1/2 & 0 \\ 0 & 1/2 & 1/2 \end{bmatrix}$$

$$P^3 = P \cdot P^2 = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1/2 & 1/2 & 0 \end{bmatrix} \begin{bmatrix} 0 & 0 & 1 \\ 1/2 & 1/2 & 0 \\ 0 & 1/2 & 1/2 \end{bmatrix} = \begin{bmatrix} 1/2 & 1/2 & 0 \\ 0 & 1/2 & 1/2 \\ 1/4 & 1/4 & 1/2 \end{bmatrix}$$

$$P^4 = P^3 \cdot P = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1/2 & 1/2 & 0 \end{bmatrix} \begin{bmatrix} 1/2 & 1/2 & 0 \\ 0 & 1/2 & 1/2 \\ 1/4 & 1/4 & 1/2 \end{bmatrix} = \begin{bmatrix} 0 & 1/2 & 1/2 \\ 1/4 & 1/4 & 1/2 \\ 1/4 & 1/2 & 1/4 \end{bmatrix}$$

$$P^5 = P \cdot P^4 = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1/2 & 1/2 & 0 \end{bmatrix} \begin{bmatrix} 0 & 1/2 & 1/2 \\ 1/4 & 1/4 & 1/2 \\ 1/4 & 1/2 & 1/4 \end{bmatrix} = \begin{bmatrix} 1/4 & 1/4 & 1/2 \\ 1/4 & 1/2 & 1/4 \\ 1/8 & 3/8 & 1/2 \end{bmatrix}$$

we observe that in P^5 all the entries are positive.
Thus P is a regular stochastic matrix.

we have to find $v = (a, b, c)$ where $a+b+c=1$ such that $vp = v$.

$$[a, b, c] \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1/2 & 1/2 & 0 \end{bmatrix} = [a, b, c]$$

$$[c/2, a + c/2, b] = [a, b, c]$$

$$c/2 = a; a + c/2 = b; b = c$$

$$\Rightarrow a + c/2 = b$$

$$a = b - c/2$$

$$a = 2a - 2a/2$$

$$c = 2a \quad \& \quad b = c = 2a \quad \text{in } a+b+c=1 \text{ we get } \Rightarrow \frac{4a-2a}{2} = \frac{2a}{2} = a$$

$$2a + 2a + a = 1$$

$$5a = 1 \Rightarrow a = 1/5 \quad \& \quad b = c = 2a \Rightarrow 2/5$$

Thus $(1/5, 2/5, 2/5)$ is the required unique fixed probability vector of P .

11) The transition matrix P of a Markov chain is given by $\begin{bmatrix} 1/2 & 1/2 \\ 3/4 & 1/4 \end{bmatrix}$ with the initial probability distribution $P^{(0)} = (1/4, 3/4)$

find the following i) $P_{21}^{(2)}$ ii) $P_{12}^{(2)}$ iii) $P^{(2)}$ iv) $P_1^{(2)}$

v) the vector $P^{(0)} P^n$ approaches

vi) the matrix P^n approaches.

i) $P_{21}^{(2)}$ is the probability of moving from state a_0 to state a_1 in 2 steps.

This can be obtained from the 2-step transition matrix P^2 .

$$P^2 = \begin{bmatrix} 1/2 & 1/2 \\ 3/4 & 1/4 \end{bmatrix} \begin{bmatrix} 1/2 & 1/2 \\ 3/4 & 1/4 \end{bmatrix} = \begin{bmatrix} 5/8 & 3/8 \\ 9/16 & 7/16 \end{bmatrix} = \begin{bmatrix} P_{11}^{(2)} & P_{12}^{(2)} \\ P_{21}^{(2)} & P_{22}^{(2)} \end{bmatrix}$$

$$\therefore P_{21}^{(2)} = 9/16$$

ii) $P_{12}^{(2)}$ is the probability of moving from state a_1 to a_0 in 2 steps

$$\therefore P_{12}^{(2)} = 3/8$$

iii) $P^{(2)}$ is the probability distribution of system in 2 steps

$$P^{(2)} = P^{(0)} P^2 = \left(\frac{1}{4}, \frac{3}{4} \right) \begin{bmatrix} 5/8 & 3/8 \\ 9/16 & 7/16 \end{bmatrix} = \left(\frac{37}{64}, \frac{27}{64} \right)$$

$$P^{(2)} = \left[\frac{37}{64}, \frac{27}{64} \right] = [P_1^{(2)}, P_2^{(2)}]$$

iv) $P_1^{(2)}$ is the probability that the process is in the state a_1 after 2 steps.

$$\therefore P_1^{(2)} = 37/64$$

v) vector $P^{(0)} P^n$ approaches the unique fixed probability.

$v = (x, y)$ where $x+y=1$ & we must have $vP = v$.

$$[x, y] \begin{bmatrix} 1/2 & 1/2 \\ 3/4 & 1/4 \end{bmatrix} = [x, y]$$

$$x/5 + 3y/4 = x \quad \text{--- (1)} \quad \& \quad x/5 + y/4 = y \rightarrow \text{(2)}$$

using $y = 1 - x$ the eqⁿ (1) becomes

$$x/5 + \frac{3(1-x)}{4} = x \quad \Rightarrow \quad \frac{2x + 3(1-x)}{4} = x$$

$$= 2x + 3 - 3x = 4x \quad \Rightarrow \quad 3 = 5x \Rightarrow \boxed{x = 3/5}$$

Since $x = 3/5$, $y = 2/5$

$\therefore$ The vector $p^{(0)}$ approaches the vector $(3/5, 2/5)$

vii) P^n approaches the matrix V whose rows are each the fixed probability vector of P .

$\therefore P^n$ approaches the matrix $\begin{bmatrix} 3/5 & 2/5 \\ 3/5 & 2/5 \end{bmatrix}$

12) The t.p.m of a markov chain is given by $\therefore$

$$P = \begin{bmatrix} 1/2 & 0 & 1/2 \\ 1 & 0 & 0 \\ 1/4 & 1/2 & 1/4 \end{bmatrix}$$

and the initial probability

distribution is $p^{(0)} = (1/2, 1/2, 0)$ find $P_{13}^{(2)}$, $P_{23}^{(2)}$ & $P_1^{(2)}$

$$P^2 = \begin{bmatrix} 1/2 & 0 & 1/2 \\ 1 & 0 & 0 \\ 1/4 & 1/2 & 1/4 \end{bmatrix} \begin{bmatrix} 1/2 & 0 & 1/2 \\ 1 & 0 & 0 \\ 1/4 & 1/2 & 1/4 \end{bmatrix} = \begin{bmatrix} 3/8 & 1/4 & 3/8 \\ 1/2 & 0 & 1/2 \\ 1/16 & 1/8 & 3/16 \end{bmatrix}$$

$$\therefore P_{13}^{(2)} = 3/8 \quad \& \quad P_{23}^{(2)} = 1/2$$

$$P^{(2)} = P^{(0)} \cdot P^2 = \begin{bmatrix} 1/2 & 1/2 & 0 \end{bmatrix} \begin{bmatrix} 3/8 & 1/4 & 3/8 \\ 1/2 & 0 & 1/2 \\ 1/16 & 1/8 & 3/16 \end{bmatrix}$$

$$= \begin{bmatrix} 7/16 & 1/8 & 7/16 \end{bmatrix}$$

$$\therefore P^{(2)} = \begin{bmatrix} 7/16 & 1/8 & 7/16 \end{bmatrix} \quad \& \quad P_1^{(2)} = 7/16$$

137. Prove that Markov chain whose t.p.m is

$$P = \begin{bmatrix} 0 & 2/3 & 1/3 \\ 1/2 & 0 & 1/2 \\ 1/2 & 1/2 & 0 \end{bmatrix} \text{ is irreducible. Find the corresponding}$$

Stationary probability vector.

First we should show that P is a regular stochastic matrix consider;

$$P^2 = \begin{bmatrix} 0 & 2/3 & 1/3 \\ 1/2 & 0 & 1/2 \\ 1/2 & 1/2 & 0 \end{bmatrix} \begin{bmatrix} 0 & 2/3 & 1/3 \\ 1/2 & 0 & 1/2 \\ 1/2 & 1/2 & 0 \end{bmatrix} = \begin{bmatrix} 1/2 & 1/6 & 1/3 \\ 1/4 & 7/12 & 7/6 \\ 1/4 & 1/3 & 5/12 \end{bmatrix}$$

Since all entries in P^2 are +ve

Therefore P is the regular stochastic matrix (or) irreducible.

Now, we should find $v = (x, y, z)$ where $x + y + z = 1$

such that; $vP = v$

$$[x \ y \ z] \cdot \begin{bmatrix} 0 & 2/3 & 1/3 \\ 1/2 & 0 & 1/2 \\ 1/2 & 1/2 & 0 \end{bmatrix} = [x \ y \ z]$$

$$= \left[\frac{y}{2} + \frac{z}{2} \quad \frac{2x}{3} + \frac{z}{2} \quad \frac{x}{3} + \frac{y}{2} \right] = [x \ y \ z]$$

$$x = \frac{y}{2} + \frac{z}{2}$$

$$x = \frac{1}{2} (y + z)$$

$$x = \frac{1}{2} (1 - x)$$

$$3x = 1 - x$$

$$3x = 1 - x \Rightarrow x = 1/3$$

$$y = 10/27 \quad \& \quad z = 8/27$$

14) A habitual is a member of two clubs A and B. He visits either of the clubs everyday for playing cards. He never visits club A on two consecutive days. But if he visits club B on a particular day, then the next day he is as likely to visit club B or club A. Find the transition matrix of this Markov chain. Also.

a) Show that the matrix is a regular stochastic matrix & find the unique fixed probability vector

b) If the person had visited club B on Monday, find the probability that he visit club A on Tuesday.

The TPM of the given Markov chain is

$$P = \begin{matrix} & \begin{matrix} A & B \end{matrix} \\ \begin{matrix} A \\ B \end{matrix} & \begin{bmatrix} 0 & 1 \\ 1/2 & 1/2 \end{bmatrix} \end{matrix}$$

$$P^2 = \begin{bmatrix} 0 & 1 \\ 1/2 & 1/2 \end{bmatrix} \begin{bmatrix} 0 & 1 \\ 1/2 & 1/2 \end{bmatrix} = \begin{bmatrix} 1/2 & 1/2 \\ 1/4 & 3/4 \end{bmatrix}$$

Since all entries in P^2 are positive, there P is the regular stochastic matrix

we should find $v = (x, y)$ where $x + y = 1 \Rightarrow x = 1 - y$

$$vP = v$$

$$(x \ y) \begin{bmatrix} 0 & 1 \\ 1/2 & 1/2 \end{bmatrix} = (x \ y)$$

$$\left[y, \quad x + y/2 \right] = [x \quad y]$$

$$x = y/2 \quad \& \quad y = x + y/2$$

$$y = 2x$$

$$y = 2(1 - y)$$

$$y = 2/3$$

$$y = 2x$$

$$2/3 = 2x$$

$$x = 1/3$$

$$\text{Thus } v = (1/3, 2/3)$$

ii) consider

$$P^3 = P \cdot P^2 = \begin{bmatrix} 1/2 & 1/2 \\ 1/4 & 3/4 \end{bmatrix} \begin{bmatrix} 0 & 1 \\ 1/2 & 1/2 \end{bmatrix} = \begin{bmatrix} 1/4 & 3/4 \\ 3/8 & 5/8 \end{bmatrix}$$

$$P_{3,1}^{(3)} = 3/8$$

let us suppose Monday as the day 1, then Thursday will be 3 days after Monday

given that person had visited club B on Monday; the probability that he visits club A after 3 days is equivalent to $P_{2,1}^{(3)}$ of P^3

thus the required probability is $P_{2,1}^{(3)} = 3/8$

15) A student's study habits are as follows. If he studies one night, he is 70% sure not to study the next night. On the other hand if he does not study one night, he is 60% sure not to study the next night. In the long run how often does he study?

The state space is $\{A, B\}$

where A is studying & B is not studying

the tpm is given by $P = \begin{matrix} & \begin{matrix} A & B \end{matrix} \\ \begin{matrix} A \\ B \end{matrix} & \begin{bmatrix} 0.3 & 0.7 \\ 0.4 & 0.6 \end{bmatrix} \end{matrix}$

we should find $V = (x, y)$ where $x + y = 1 \Rightarrow y = 1 - x$ such that $VP = V$

$$\begin{bmatrix} x & y \end{bmatrix} \begin{bmatrix} 0.3 & 0.7 \\ 0.4 & 0.6 \end{bmatrix} = \begin{bmatrix} x & y \end{bmatrix}$$

$$x = 0.3x + 0.4y \quad \& \quad y = 0.7x + 0.6y$$

$$0.3x + 0.4(1-x) = 0$$

$$\boxed{x = 4/11}$$

$$y = 1 - \frac{4}{11}$$

$$\boxed{y = 7/11}$$

$\therefore$ Thus $V = (4/11, 7/11)$ is the unique fixed p.v.

$$\therefore \left(\frac{4}{11}, \frac{7}{11} \right) \cdot (P_A, P_B)$$

$\therefore$ Thus the prob of studying in long run is $4/11$

16) A Man's smoking habits are as follows. If he smokes filter cigarettes one week, he switches to non-filter cigarette the next week with probability 0.3. On the other hand, if he smokes non-filter cigarettes one week there is a probability of 0.7 that he will smoke non-filter cigarettes the next week as well. In the long run how often does he smoke filter cigarettes?

The space of the system is $\{A, B\}$ where A is smoking filter cigarettes & B is smoking non filter cigarettes.

$$P = \begin{matrix} & \begin{matrix} A & B \end{matrix} \\ \begin{matrix} A \\ B \end{matrix} & \begin{bmatrix} 0.8 & 0.3 \\ 0.3 & 0.7 \end{bmatrix} \end{matrix} = \begin{bmatrix} 8/10 & 3/10 \\ 3/10 & 7/10 \end{bmatrix} = \frac{1}{10} \begin{bmatrix} 8 & 3 \\ 3 & 7 \end{bmatrix}$$

$$v = (x, y) \quad \& \quad x + y = 1, \quad vP = v.$$

$$[x \quad y] \cdot \frac{1}{10} \begin{bmatrix} 8 & 3 \\ 3 & 7 \end{bmatrix} = [x \quad y]$$

$$[8x + 3y, 3x + 7y] = [10x, 10y]$$

$$8x + 3y = 10x \quad \text{--- (1)} \quad , \quad 3x + 7y = 10y \quad \text{--- (2)}$$

$$y = 1 - x \quad \text{--- (1) becomes.}$$

$$8x + 3(1-x) = 10x$$

$$\boxed{x = 3/5} \quad \& \quad \boxed{y = 2/5}$$

$$\text{Hence } v = (x, y) = (3/5, 2/5) = (p_A, p_B).$$

In the long run, he will smoke filter cigaret is $3/5$

17) Each year a man trades his car for a new car in 3 brands of the popular company Maruti udyog limited. If he has a 'standard' he trades it for 'zen'. If he has a 'zen' he trades it for a 'Esteem'. If he has a 'Esteem' he is just as likely to trade it for a new 'Esteem' or for a 'zen' or a 'standard' one. In 1996 he bought his first car which was Esteem

i) find the probability that he has

a) 1998 Esteem b) 1998 Standard

c) 1999 Zen d) 1999 Esteem

ii) In the long run, how often will he have a Esteem?

we know ^{have} state space = $\{A, B, C\}$

where A - standard, B - zen, C - Esteem

the TPM is given by

$$P = \begin{matrix} & \begin{matrix} A & B & C \end{matrix} \\ \begin{matrix} A \\ B \\ C \end{matrix} & \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1/3 & 1/3 & 1/3 \end{bmatrix} \end{matrix} = \begin{bmatrix} P_{11} & P_{12} & P_{13} \\ P_{21} & P_{22} & P_{23} \\ P_{31} & P_{32} & P_{33} \end{bmatrix}$$

$$i) P^2 = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1/3 & 1/3 & 1/3 \end{bmatrix} \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1/3 & 1/3 & 1/3 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 1 \\ 1/3 & 1/3 & 1/3 \\ 1/9 & 4/9 & 4/9 \end{bmatrix}$$

$$P^3 = \begin{bmatrix} 0 & 0 & 1 \\ 1/3 & 1/3 & 1/3 \\ 1/9 & 4/9 & 4/9 \end{bmatrix} \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1/3 & 1/3 & 1/3 \end{bmatrix} = \begin{bmatrix} 1/3 & 1/3 & 1/3 \\ 1/9 & 4/9 & 4/9 \\ 4/27 & 7/27 & 16/27 \end{bmatrix}$$

$$a) 1998 \text{ Esteem} = P_{33}^2 = 4/9$$

$$b) 1998 \text{ Standard} = P_{31}^2 = 1/9$$

$$c) 1999 \text{ Zen } P_{32}^3 = 7/27$$

$$d) 1999 \text{ Esteem } P_{33}^3 = 16/27$$

ii) we should find $v = (x, y, z)$ where $x+y+z=1$
 $\Rightarrow z = 1-x-y$

such that $vP = v$

$$[x \ y \ z] \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1/3 & 1/3 & 1/3 \end{bmatrix} = [x \ y \ z]$$

$$\begin{bmatrix} x & x + z/3 & y + z/3 \end{bmatrix} = [x \ y \ z]$$

$$x = z/3 \quad \& \quad y = x + z/3 \quad \& \quad z = y + z/3$$

$$z = 3x \quad ; \quad 3y = 3x + z \quad ; \quad 3z = 3y$$

solving these eqns by using $x + y + z = 1$

$$3x + z = 3y$$

$$3x + z = (1 - x - z)3$$

$$6x + 4z = 3$$

$$6x + 4(3x) = 3$$

$$\boxed{x = 1/6}$$

$$z = 3x$$

$$z = 3(1/6)$$

$$\boxed{z = 1/2}$$

$$3z = 3y$$

$$3(1/2) = 3y$$

$$\boxed{y = 1/3}$$

thus $v = (1/6, 1/3, 1/2) = (p_A, p_B, p_C)$ is the unique fixed PV

$\therefore$ In the long run, ~~prob~~ prob of having Esteen is. $p_C = 1/2$